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UNIVERSITÀ DI PADOVA
Dipartimento
di Biologia

Keeping it short, going long, and trusting Sanger

From Sanger sequencing to NGS

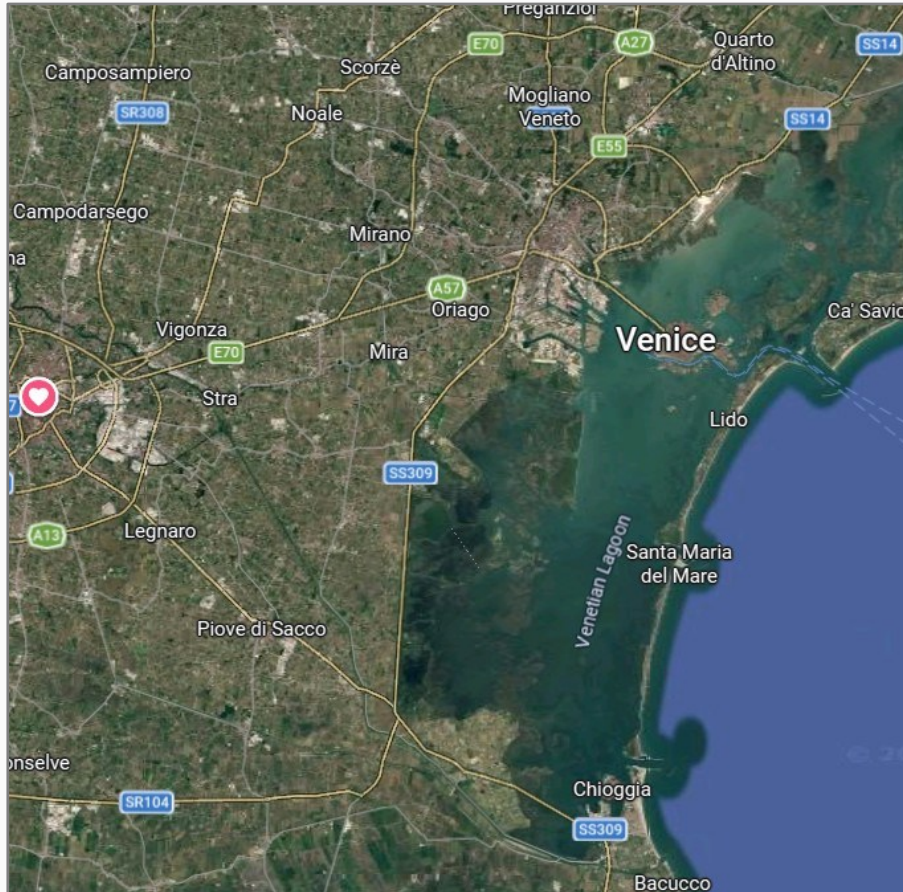
Alessandro Vezzi,
Molecular biologist, PhD in biotechnology

Where is Padova? And what about its University?

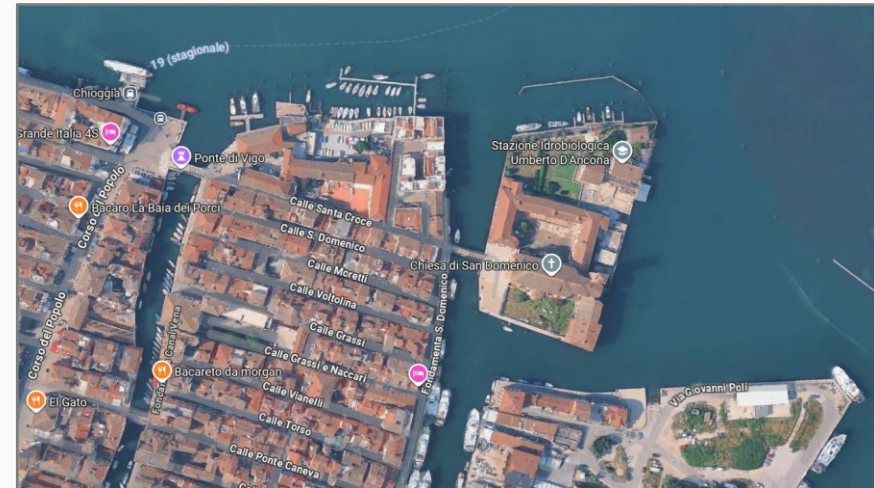


- Small town, large student population
- Top University in Italy
- 32 different Departments
- Various external venues

One external venue is Chioggia



- Hydrobiological station
- Direct access to lagoon and sea



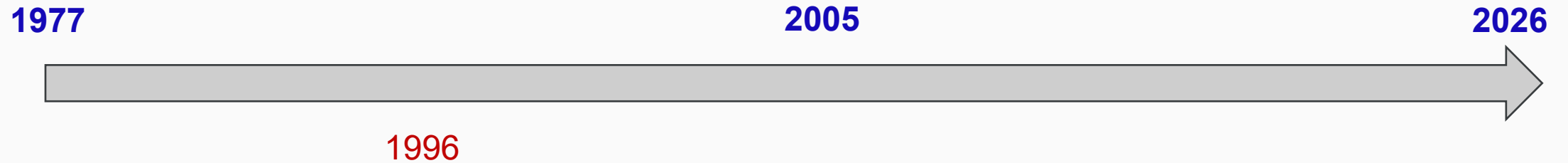
And what about me?

- From an island of Venice
- Wet lab oriented
- Runner
- Mountain lover



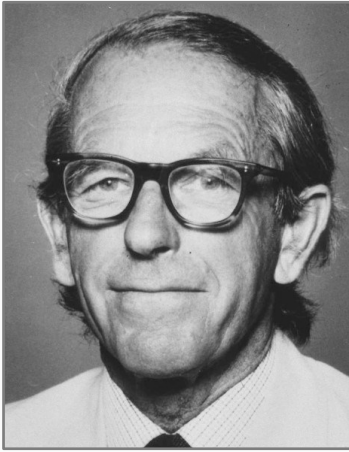
Why am I here?

Keeping it **short**, going **long**, and **trusting** Sanger
From Sanger sequencing to NGS



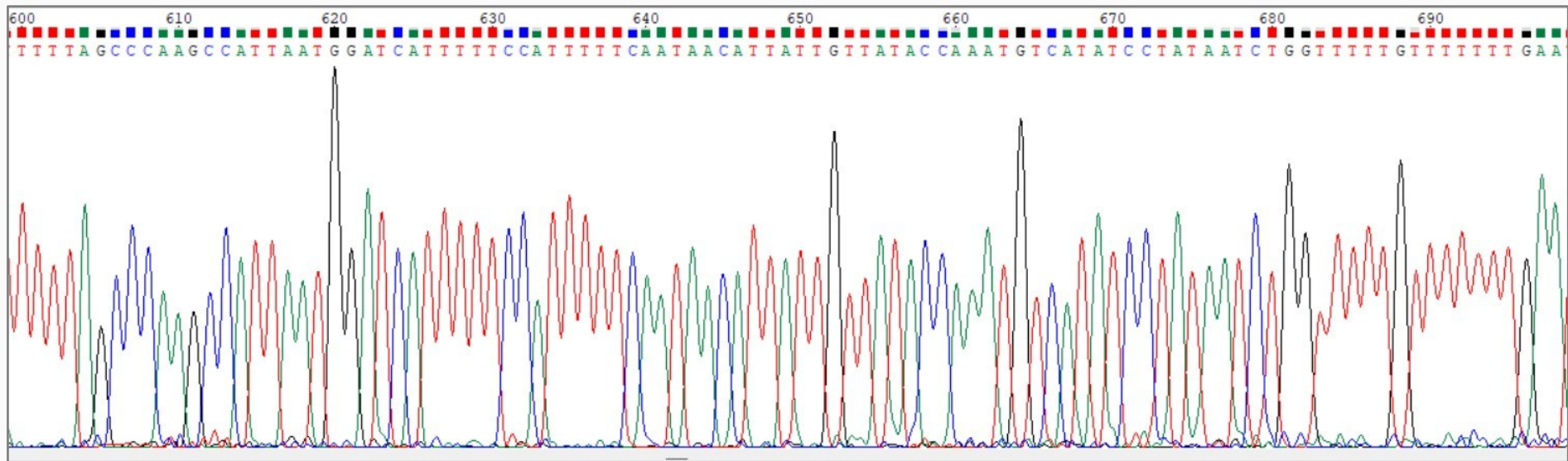
Both Sanger sequencing and NGS have evolved over the years

- Upstream laboratory protocols
- Read length
- Error rate per base
- Instruments
- Throughput
- Dedicated software
- Increased computational needs

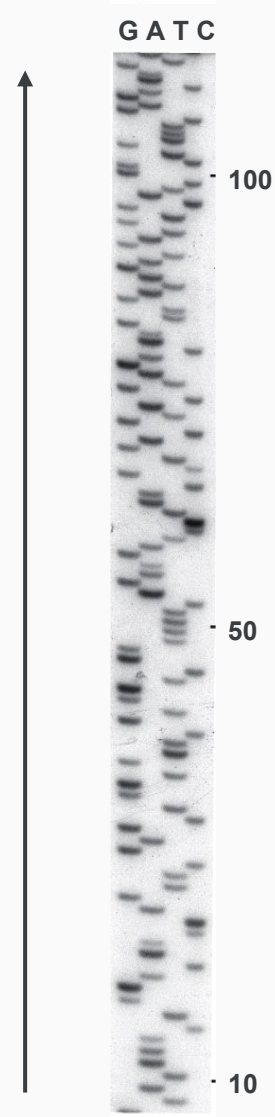


Sanger sequencing, now

- Different fluorescent dyes for each nucleotide
- Good quality for “many” bases



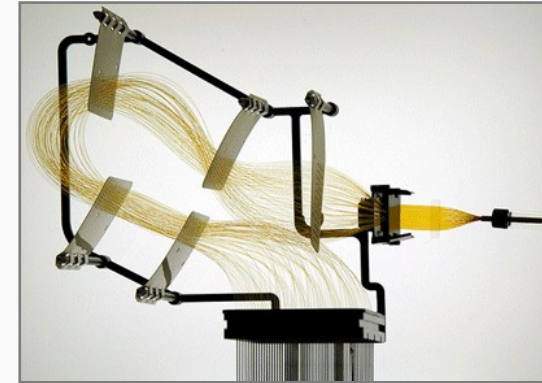
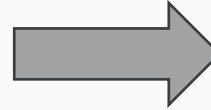
Sanger sequencing, then



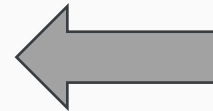
- Time-consuming protocols
- Radioactively labelled nucleotides
- Four reactions for each sample
- Manual call of the base order (highly prone to **errors**)

From manual to automatic Sanger sequencing

- Polyacrylamide gel electrophoresis
- Southern blot and autoradiography
- Manual call of the base order



- Around **1 Mb/day** per instrument
- Around **99.99% accuracy**



Thus, **we trust** Sanger sequencing



Sanger technology limitations

- **Low throughput**
($\approx$ 5 days for **1X coverage** of a 5 Mbp bacterial genome)
- **Quite expensive** per base
- **Labor-intensive** (i.e., cloning steps, colony collection)
(2500 clones with an average insert size of 2kbp to obtain a 1X coverage of a 5Mbp genome (**very rough estimation**))

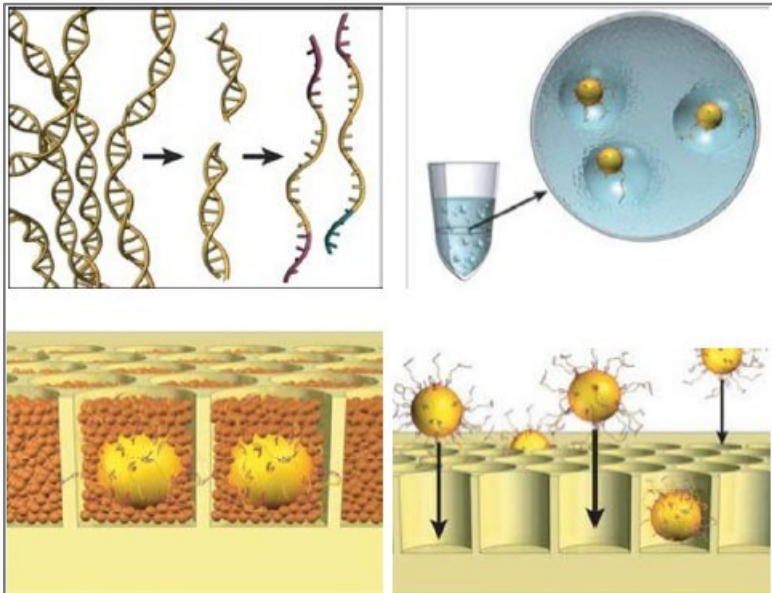
Still,

Tyson, G. *et al.* (2004) *Nature* 428, 37–43. <https://doi.org/10.1038/nature02340>

Rusch, D.B. *et al.* (2007) *PLoS Biol* 5(3): e77. <https://doi.org/10.1371/journal.pbio.0050077>

Then, the NGS era arrived

454: Margulies, M. *et al.* (2005) Genome sequencing in microfabricated high-density picolitre reactors. *Nature* 437, 376–380. <https://doi.org/10.1038/nature03959>



- 25 Mb in one four-hour run
- Around 100-fold throughput increase
- Average read accuracy > 99%
- Average read length: 110 bases



Here we are with the **short reads**

Short read sequencing platforms

- Defined **second generation** sequencers
 - No subcloning needed
 - PCR amplification still needed (**signal amplification**)
 - Each read is obtained from the combined signal of “identical copies” of the same DNA fragment
- The PCR amplification step **can introduce errors**, which vary depending on the adopted sequencing platform.

Different short read sequencing platform are (were) available (Non-exhaustive list)



454



Illumina



Solid



Ion Proton

With different

- Upstream laboratory protocols
- Sequencing methods (synthesis vs ligation)
- Signal detection methods (pH changes, fluorescence, light emission)

which mean

- Keeping pace with technological developments
- Relying on different data analysis software (sometimes customised)
- Too expensive for a small/medium-sized laboratory: **outsourcing**

Some platforms become obsolete very quickly



Our SOLiD system in 2016

Other platforms vary greatly.1

Benchtop sequencers

[Learn how to choose a benchtop sequencer](#)

			
Key specifications	<u>MiSeq i100 Series</u>	<u>NextSeq 550 System</u>	<u>NextSeq 1000 and 2000 Systems</u>
Max output per flow cell	30 Gb ^a	120 Gb ^b	540 Gb
Run time (range) ^c	~4–24 hr	~11–29 hr	~8–44 hr
Max reads per run (single reads)	100M ^a	400M ^b	1.8B
Max read length	2 × 500 bp	2 × 150 bp	2 × 300 bp

Other platforms vary greatly.2

Production-scale sequencers

Key specifications	 <u>NextSeq 1000 and 2000 Systems</u>	 <u>NovaSeq 6000 System</u>	 <u>NovaSeq X Series</u>
Max output per flow cell	540 Gb ^a	3 Tb ^b	8–10.5 Tb ^c
Run time (range) ^d	~8–44 hr	~13–44 hr	~17–48 hr
Max reads per run (single reads)	1.8B ^a	10B (single flow cell) ^b 20B (dual flow cells)	26–35B (single flow cell) ^c 52–70B (dual flow cells) ^{c,e}
Max read length	2 × 300 bp	2 × 250 bp	2 × 300 bp

<https://emea.illumina.com/systems/sequencing-platforms.html>

Some possible drawbacks?

- Take time to completely fill the flow cell
- High throughput: high computational power
- Generally, short reads

Does the read length really matter?

It depends on the research you wish to carry out

- **Metabarcoding and eDNA studies**

If the theoretical length of the **merged** reads > the length of the amplicons: **no matter**
(i.e., 2*250 bp is theoretically sufficient for a V4-V5 16S rRNA amplicon library)

- **Genomics/transcriptomics/metagenomics**

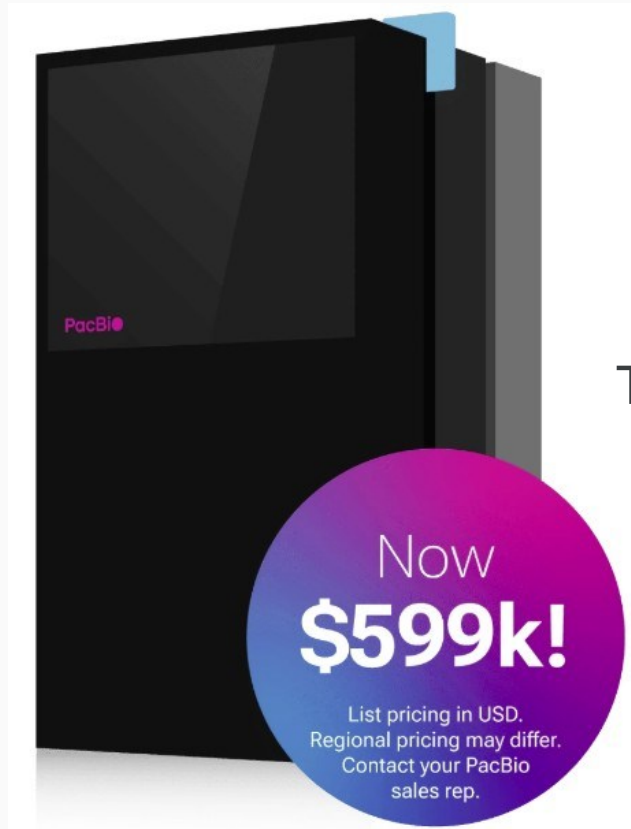
What is your biological question?

- 1) Trying to characterise the functional potential of a microbial community? **No matter**
- 2) Trying to untangle the highly repetitive genome of a eukaryotic organism? **Lots of problem**
- 3) Trying to define the different isoforms of a *de novo* transcriptome? **Again, lots of problem**
- 4) ...

Long reads sequencing platforms

- Defined **third generation** sequencers
 - No subcloning needed
 - No PCR amplification*: single-molecule sequencing/reading
 - Real time sequencing/reading
 - Long reads
 - full rRNA operon (metabarcoding)
 - chromosome level genome (genomics)
 - isoform resolution (transcriptomics)

Long reads sequencing platform available



PacBio Revio

The tough ones



Oxford Nanopore PromethION 24

Some possible drawbacks?

- Lower throughput than short read sequencers
- High error rate (Nanopore), but improving
- Too expensive: **outsourcing**

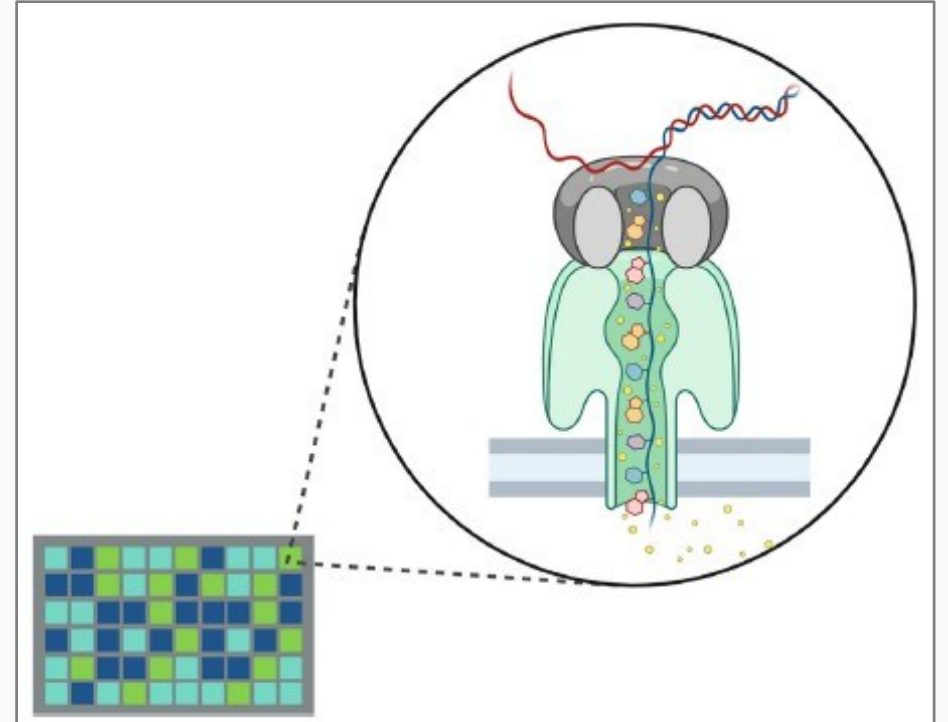
But Oxford Nanopore offers a **portable** sequencing device

Good for —————→ metabarcoding
 —————→ small/medium size genomes
 —————→ *de novo* transcriptome



Oxford Nanopore **does not synthesise** DNA

- Nanopores are embedded in a membrane
- Electric current flows through each pore
- DNA/RNA strands are translocated through each pore
- The passage of the bases (groups of bases) alters the current in a characteristic way.
- The device records the current changes



Base interpretation relies on **training data** linking current changes to bases

Final notes on Oxford Nanopore

- Technology, protocols and software in **constant development**
- If you have your own instrument, **plan sequencing carefully**
- Flow cells can be reused (but change the experiment!)
- There is an **active user community**

Few conclusions

NGS platforms have a much lower cost per base than Sanger sequencing

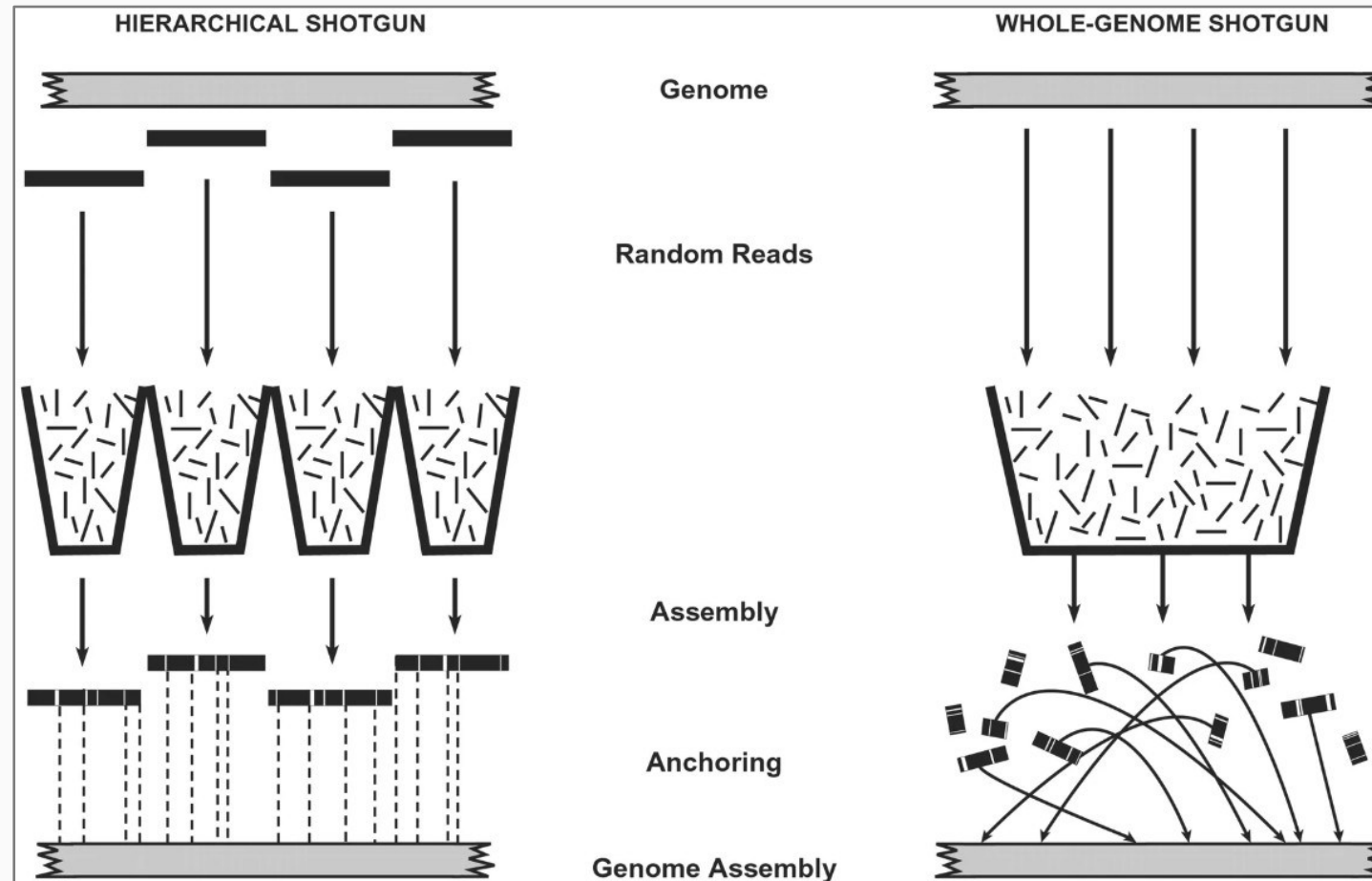
Cost per base decreases as the instrument's throughput increases (but outsourcing)

Depending on the biological question and current knowledge, choose the most appropriate sequencing platform.

What about Sanger sequencing?

Remembering John Craig Venter

The first to propose the possibility of sequencing genomes using a **whole-genome shotgun approach**



<https://doi.org/10.1073/pnas.042692499>